

AI-Powered HMS Synthetic Clinical Dataset

Lab Result

Patient Name	Bob Synthetic	MRN	MRN-0002
DOB	1969-09-09	Gender	Male
Encounter Date	2023-02-04	Department	Cardiology
Attending Provider	BS. Le Hoang Phuc	Status	active

Synthetic Data Warning: This document is generated for software testing, OCR parsing, chunking, embedding, access-control validation, and Graph RAG demonstrations. It is not valid medical advice and must not be used for patient care.

Lab Report Summary

Six-month lab trend reviewed for Bob Synthetic with emphasis on renal function, glycemic control, BNP, liver enzymes, electrolytes, and hemoglobin. Clinical context: acute on chronic heart failure with preserved ejection fraction, atrial fibrillation, hypertension.

Recent Lab Snapshot

Date	Analyte	Value	Unit	Reference Range	Status
2026-05-02	Creatinine	1.74	mg/dL	0.7-1.3	High
2026-05-02	eGFR	58.0	mL/min/1.73m2	>=60	Low
2026-05-02	Glucose	143.0	mg/dL	70-99	High
2026-05-02	HbA1c	7.8	%	<5.7	High
2026-05-02	BNP	377.0	pg/mL	<100	High
2026-05-02	AST	36.0	U/L	10-40	Normal
2026-05-02	ALT	40.0	U/L	7-56	Normal
2026-05-02	Potassium	4.5	mmol/L	3.5-5.0	Normal
2026-05-02	Hemoglobin	13.8	g/dL	12.0-17.5	Normal

Interpretation

Abnormal values should be correlated with symptoms, medications, hydration status, and department-specific care goals. High BNP or low eGFR should trigger clinician review before medication changes.

Trend Notes For AI Extraction

Renal Labs

Creatinine and eGFR are included for renal dosing and medication safety retrieval. Monitor K when ACEI, ARB, spironolactone, or diuretics are active.

Medication Safety Linkage

The lab trend is intended to link to medication_safety and renal_labs retrieval scopes. Pharmacist review is recommended for high-risk drug-lab combinations.